



# Operating Systems

[1500WETOPS]

José Oramas



# Introduction

## [ Some Basic Concepts ]

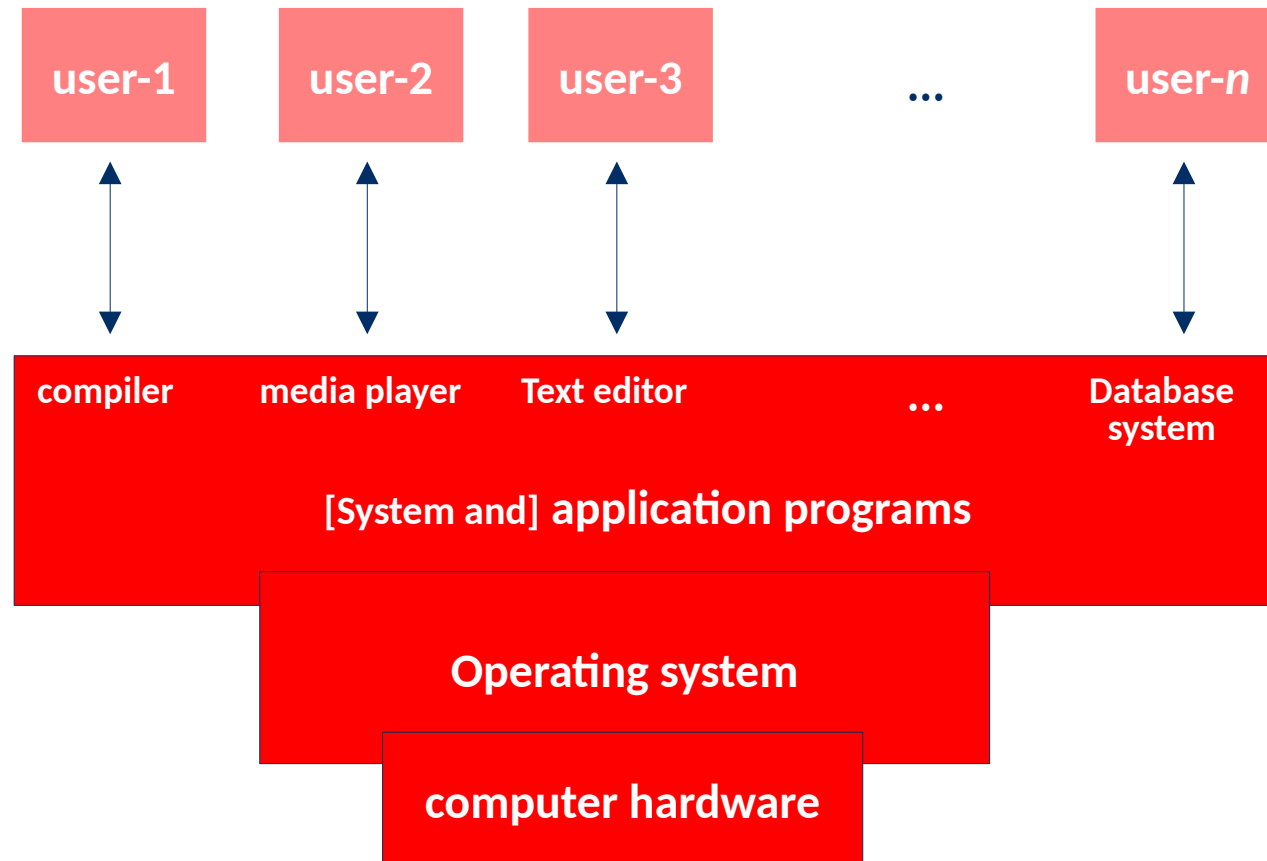
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First of all...  
**How would you define an  
operating system?**



# Introduction

## A Computer System - an abstract view



### Operating System

- Act as an intermediary [ users & hardware ]
- Provide an environment for proper use of programs.
- Convenience vs. Efficiency

# Operating Systems

## 1- A User Perspective



- **Scenario:** single-user
- Designed for the monopolization of its resources.
- High Performance, Low resource-utilization



# Operating Systems

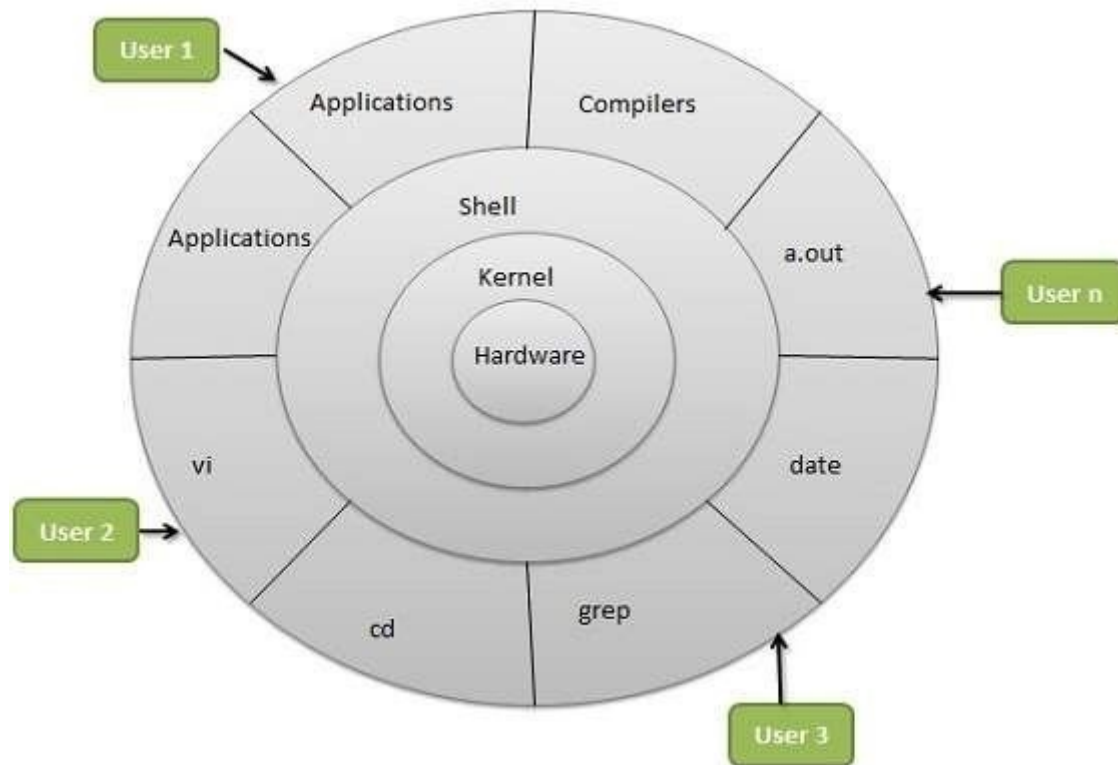
## 1- A User Perspective



- **Scenario:** multi-user
- Resources are shared among users.
- High resource-utilization, Low Performance.

# Operating Systems

## 2- A System Perspective



- Manager of resources.
  - Ensure efficient and fair operation.
- Control program
  - Execution of Programs  
[ Prevention of Errors and improper use ]
  - Operation/control of I/O devices

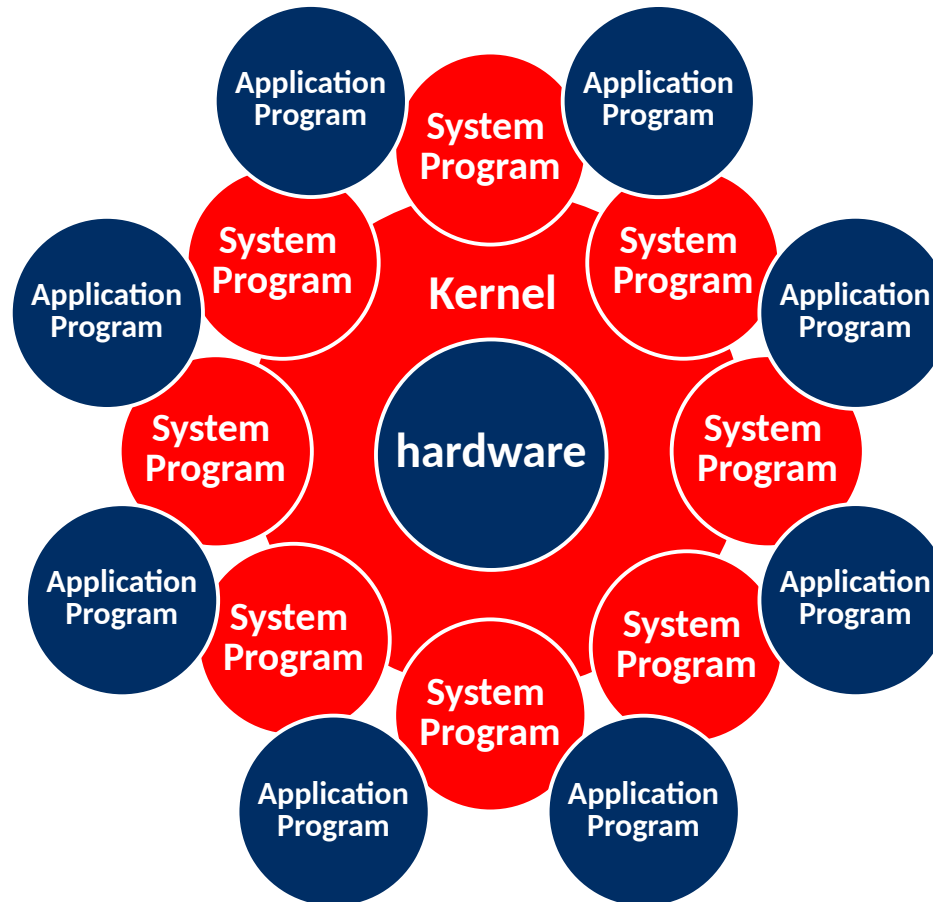
So...  
**What is then the  
proper definition?**





# Operating systems

A common definition: “The program running at all times on the computer”  
[→ kernel]



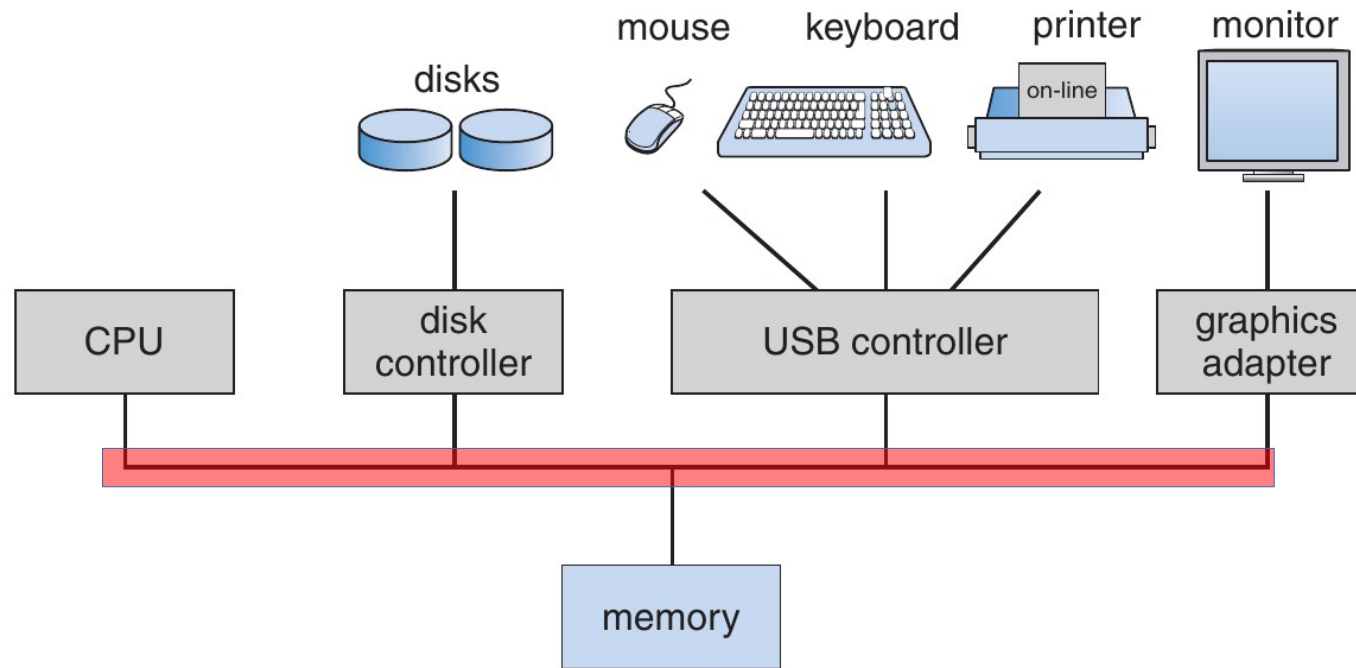
- Systems programs are associated to the OS.
- Application programs are not.
- What functionality defines a system program is not clear.



[link]

# Operating systems

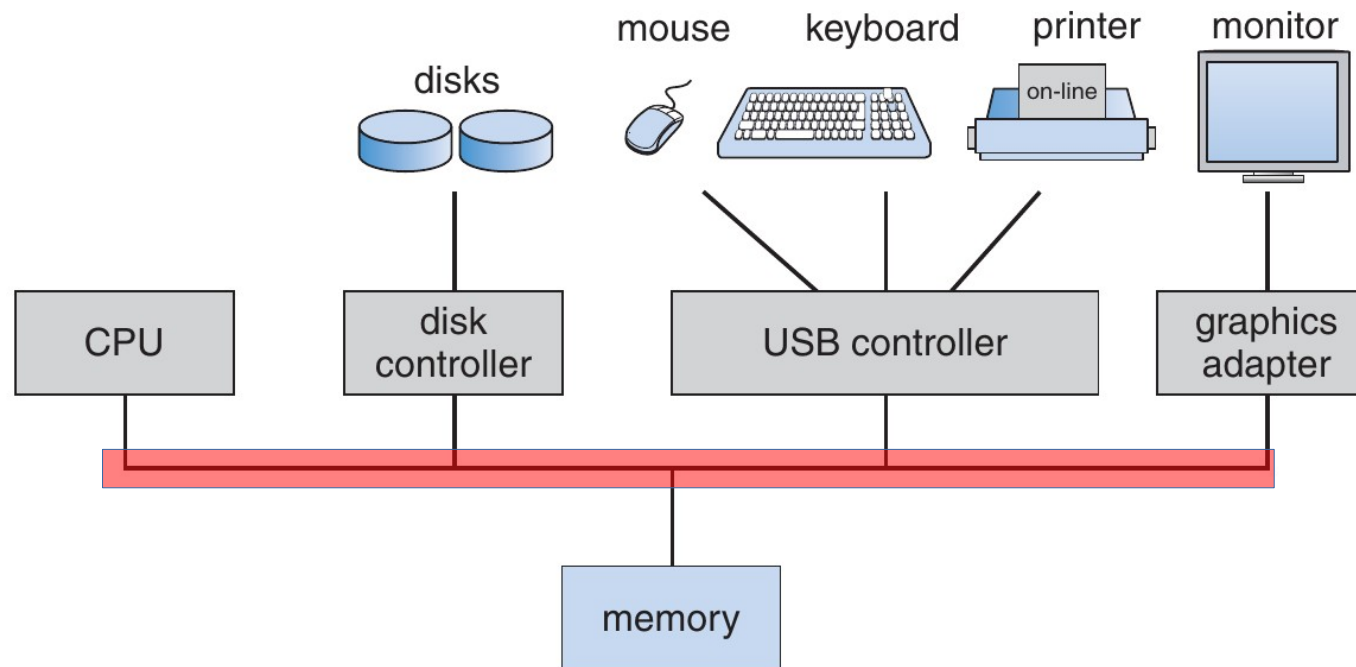
## A Modern Computer System



# Operating systems

## A Modern Computer System

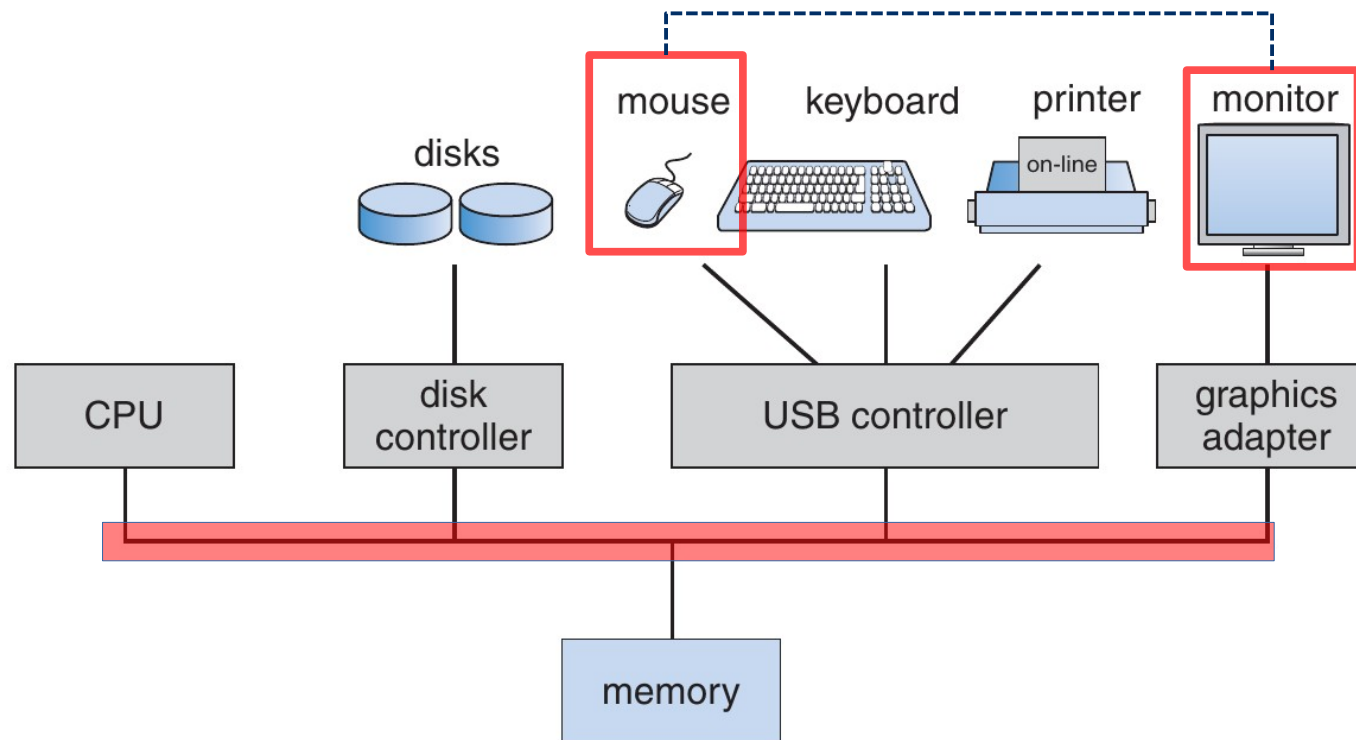
- The flow of a program execution → let's reproduce a video file



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## A Modern Computer System

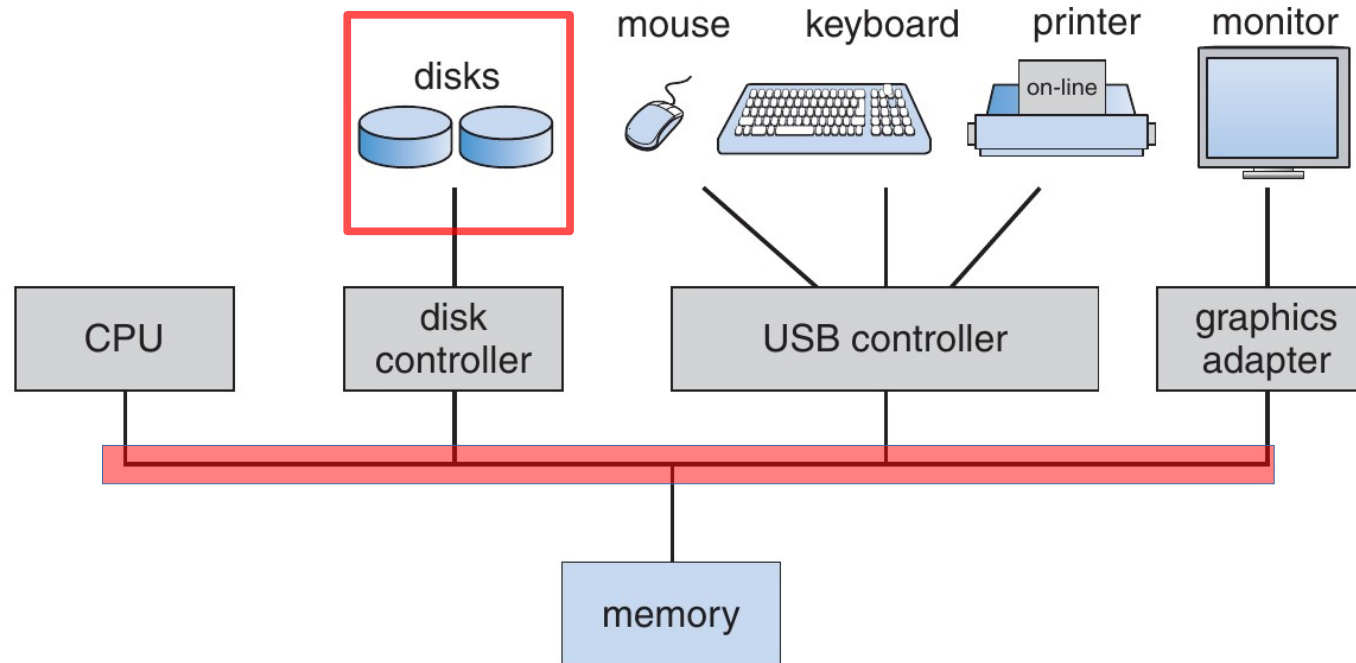
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## A Modern Computer System

- The flow of a program execution → let's reproduce a video file

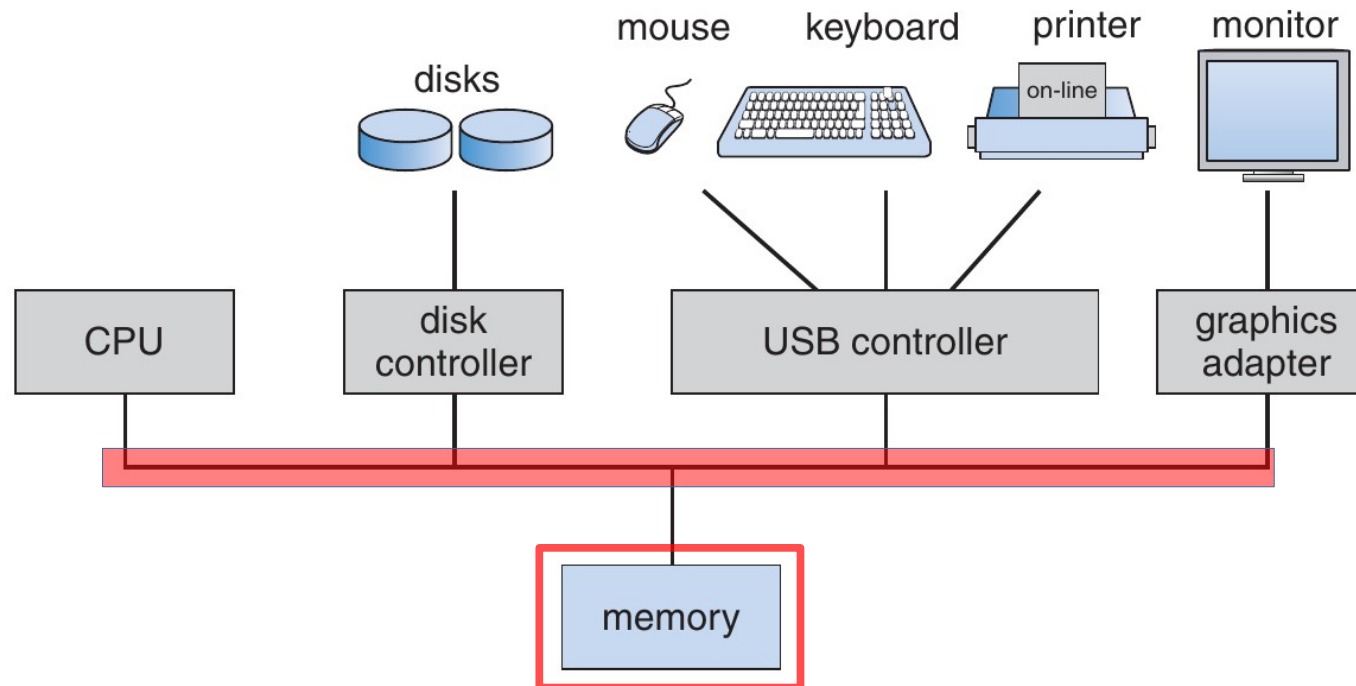




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## A Modern Computer System

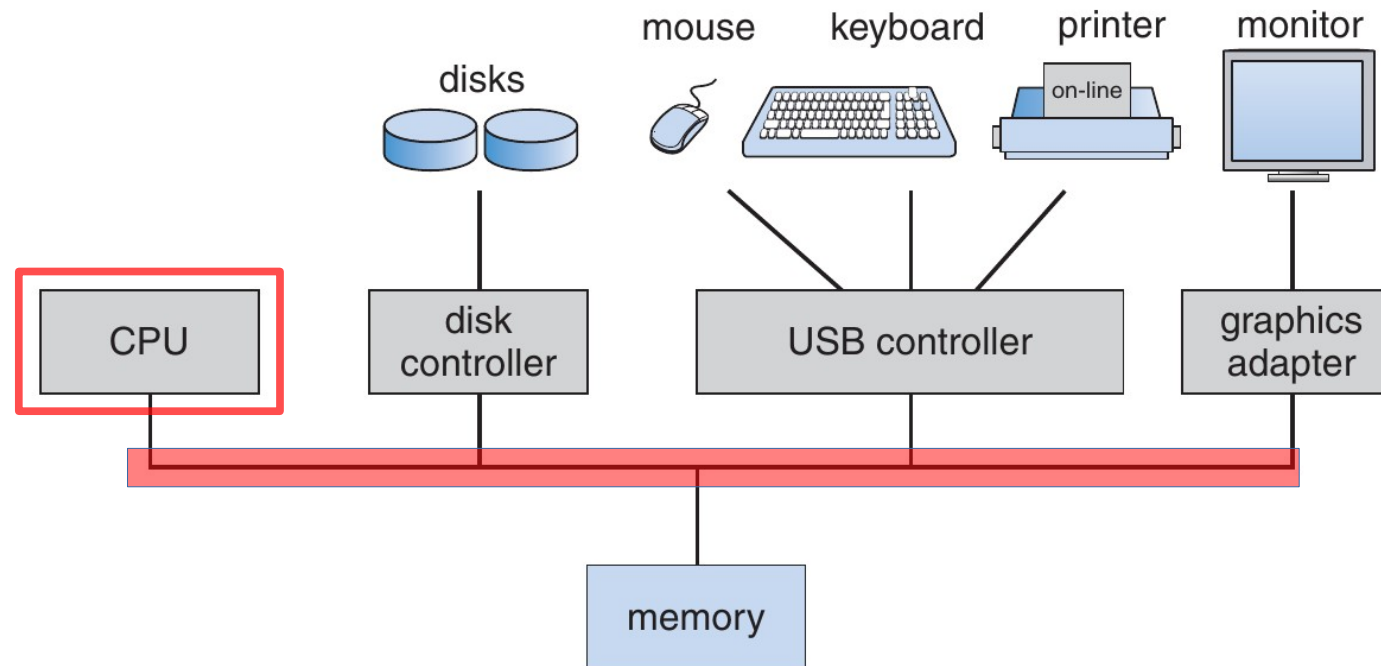
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## A Modern Computer System

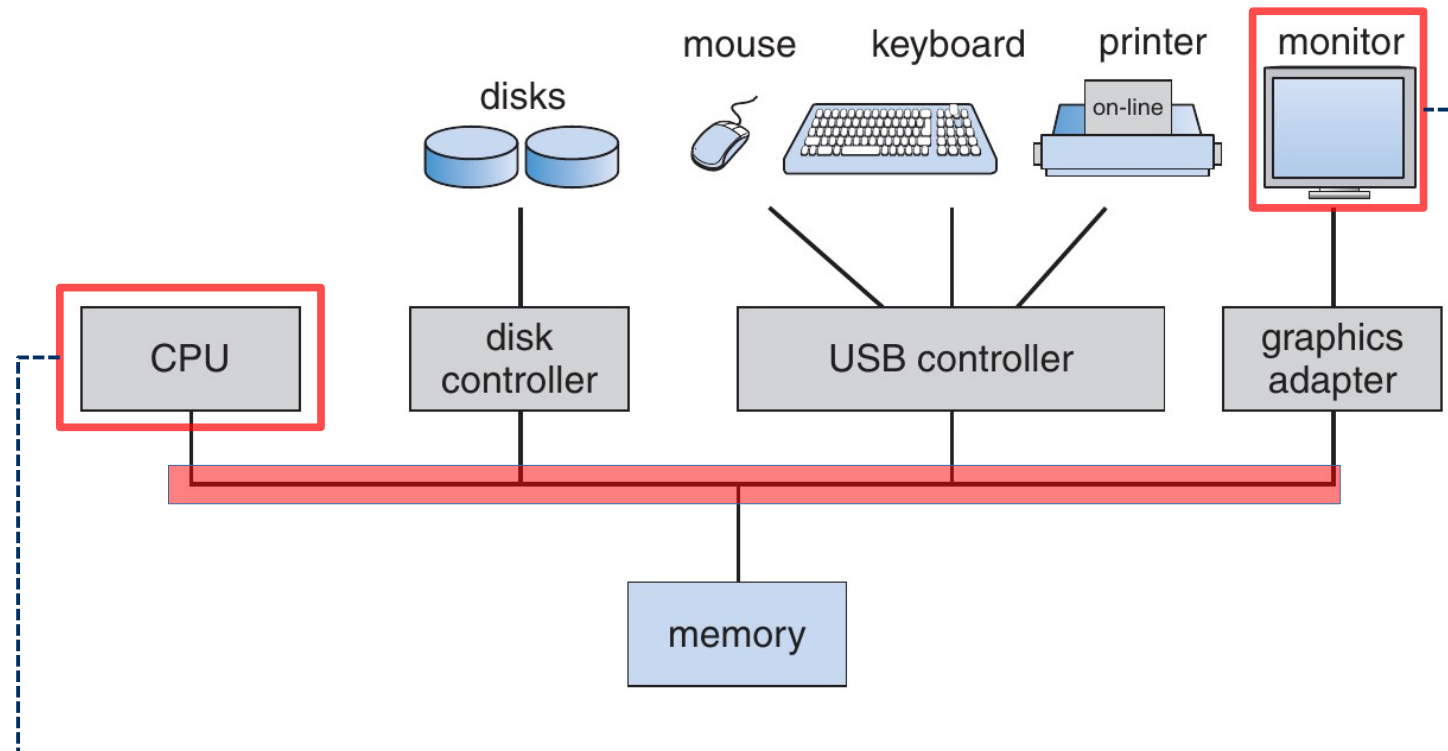
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# Operating systems

## A Modern Computer System

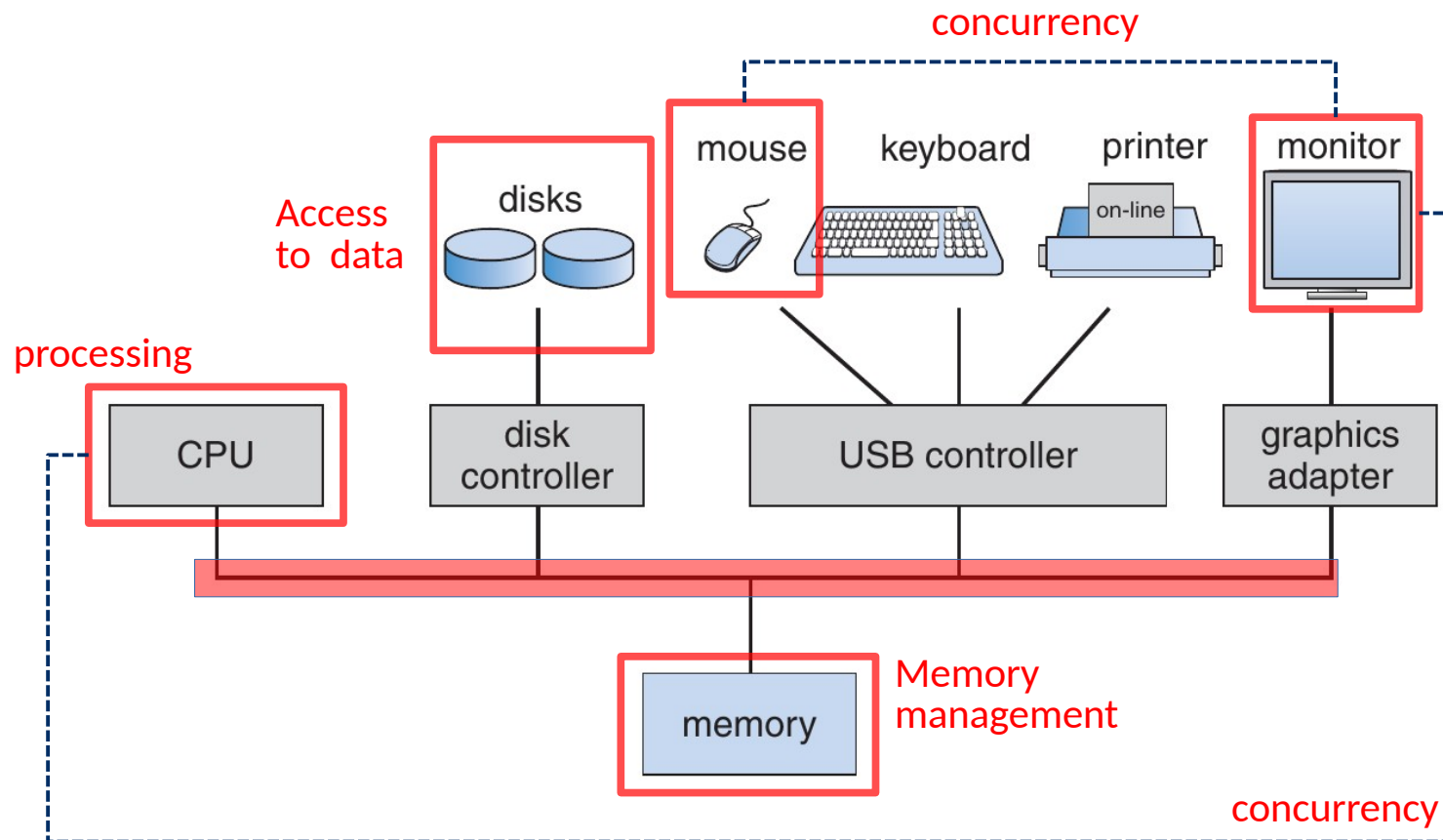
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# Operating systems

## A Modern Computer System

- The flow of a program execution → let's reproduce a video file



# Questions?

# Further Reading

## General OS / Computer Organization Introduction

- Silberschatz et al., Operating System Concepts Essentials, 2nd Edition., **Chapter-1** – *recommended*



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